Egg Segmentation and Multi-Object Tracking Take-Home Report

Candidate: Sree Varun Aahil | **Date:** October 30, 2026 | **Machine:** Windows 11 (CUDA Enabled)

0.4018	0.3196	0.4087	6	110
DEV MEAN IOU	EVAL MEAN IOU	DEV DETECTION F1	VIDEO UNIQUE TRACKS	CLEAN GROUND TRUTHS

1. Executive Summary

This report presents an end-to-end computer vision solution for egg segmentation and multi-object tracking under challenging conditions, including touching instances, boiling water, high-contrast table textures, and hand occlusions. Ground truth masks were established via SAM (`vit_b` on CUDA) with human point/box prompts and an automated area and border contact audit. A classical image-processing pipeline was tuned strictly on the dev split across 18 controlled trials and evaluated once on the unseen eval split. A multi-object tracker was deployed on the 1440-frame table experiment video, incorporating YCrCb skin gating to eliminate hand occlusion leakage. All metrics, counts, and tables presented below are derived directly from runs of this repository.

- **Ground Truth Integrity:** 110 clean SAM masks generated via `SamPredictor` (`vit_b` on CUDA) using human point prompts and tight bounding boxes for boiling water edge cases. 0 flagged suspicious masks remaining.
- **Classical Pipeline (Dev Split, n=88):** Pixel Mean IoU = **0.4018** (95% CI: [0.3435, 0.4603], Median = 0.3147), Dice = **0.5200**, Detection F1 = **0.4087**, Mean Count Error = **2.20**.
- **Classical Pipeline (Eval Split, n=22):** Pixel Mean IoU = **0.3196** (95% CI: [0.1978, 0.4268], Median = 0.2502), Dice = **0.4239**, Detection F1 = **0.3206**, Mean Count Error = **1.68**.
- **Video Tracking (1440 frames):** 1440 frames processed at 24.0 FPS. **6 unique tracks** created across the video, with **18 loss events** and **12 recoveries**. Hand occlusion and fingertip leakage at frame 600 resolved using YCrCb skin gating and morphological opening (gating reduced the total foreground mask by 2296 px over the whole frame; the fingertip bulge is visible in Figure 6, and restricted to the crop x in [200, 460], y in [300, 430], the mask was reduced by 1304 px).

2. Task 1: Ground Truth Mask Generation (SAM)

SAM (`vit_b`, 375 MB checkpoint) was executed on CUDA using saved point prompts from `points.json` with multi-mask score selection (`multimask_output=True`). Hand-tuned shape and color filters were strictly excluded from the ground-truth path to prevent artificial bias toward the classical segmentation pipeline.

Ground Truth Egg Distribution across 110 images: 93 images (84.5%) with 1 egg; 6 images (5.5%) with 2 eggs; 11 images (10.0%) with 3 eggs.

Ground Truth Quality Audit & Fixes (`out/sam/gt_fix_audit.json`)

Initial automated SAM passes on boiling water images flooded the pot interior. An automated border and area audit in `src/contact_sheet.py` flagged masks with area > 40% or edge contact > 50%. Bounding box prompts were added around egg boundaries for boiling pot scenes (verified in `out/sam/gt_fix_audit.json`):

- `not_damaged_124`: BBox `[280, 150, 520, 400]` reduced area fraction from 81.9% (335354 px, 978 border contacts) to **13.4%** (54995 px, 0 border contacts).
- `not_damaged_135`: BBox `[270, 170, 520, 420]` reduced area fraction from 79.5% (325584 px, 994 border contacts) to **14.2%** (58113 px, 0 border contacts).
- `not_damaged_145`: BBox `[250, 200, 480, 450]` reduced area fraction from 62.5% (256010 px, 665 border contacts) to **13.2%** (54032 px, 0 border contacts).

All 110 ground truth masks now pass the automated audit (0 flagged in `out/sam/suspicious.txt`).

3. Task 2: Parameterized Classical Segmentation & Experiment Log

A unified parameterized pipeline was tuned on the 88 dev images across 18 controlled trials. Every trial varied exactly one parameter relative to its named parent. Keep/discard decisions compare strictly against the parent's dev IoU. Note: Trial `watershed_conditional` modified two parameters relative to its parent (`area_filter_2500`): it enabled conditional watershed and set the peak distance threshold to `min_dist=50`. Under this trial, Detection F1 rose from 0.3084 to 0.4087, mean signed count error rose from +0.45 to +2.20 (over-splitting), pixel IoU unchanged. Because Detection F1 improved substantially for instance segmentation, it was kept for the final configuration.

Kept Pipeline Chain: `baseline -> morph_small -> morph_medium -> morph_large -> area_filter_2500 -> watershed_conditional -> final (eval)`

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`baseline`	`None`	dev	Grayscale conversion + Gaussian blur (5x5) + Otsu threshold + hole fill + min area 500	0.3513	0.4693	0.2071	3.12	BASELINE	Baseline established on dev (IoU 0.3513, Dice 0.4693, DetF1 0.2071)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`otsu_inv`	`baseline`	dev	Invert Otsu thresholding	0.0823	0.1424	0.0000	3.36	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0823 vs 0.3513, -0.2690)
`clahe_otsu`	`baseline`	dev	CLAHE local contrast equalization (clip 2.0, tile 8x8) before Otsu	0.2562	0.3791	0.0799	8.22	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.2562 vs 0.3513, -0.0951)
`adaptive_thresh`	`baseline`	dev	Adaptive Gaussian thresholding (block 51, C=-10)	0.0634	0.1119	0.0065	12.33	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.0634 vs 0.3513, -0.2878)
`lab_l`	`baseline`	dev	CIELAB L* perceptual lightness channel + Otsu	0.3244	0.4437	0.1665	3.67	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3244 vs 0.3513, -0.0268)
`hsv_gate`	`baseline`	dev	HSV color gating (S < 70, V > 130)	0.3456	0.4665	0.1587	4.32	DISCARD	Fails to improve dev IoU over parent 'baseline' (0.3456 vs 0.3513, -0.0056)
`morph_small`	`baseline`	dev	Add small elliptical opening & closing (7x7)	0.3558	0.4740	0.2161	2.89	KEEP	Improves dev IoU over parent 'baseline' from 0.3513 to 0.3558 (+0.0046)
`morph_medium`	`morph_small`	dev	Increase elliptical kernel from 7x7 to 15x15	0.3766	0.4964	0.2363	2.34	KEEP	Improves dev IoU over parent 'morph_small' from 0.3558 to 0.3766 (+0.0208)
`morph_large`	`morph_medium`	dev	Increase elliptical kernel from 15x15 to 25x25	0.3972	0.5161	0.2642	1.25	KEEP	Improves dev IoU over parent 'morph_medium' from 0.3766 to 0.3972 (+0.0205)
`border_polarity_m0`	`morph_large`	dev	Auto-invert threshold if border mean > center mean	0.3698	0.4821	0.2409	1.07	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3698 vs 0.3972, -0.0274)
`border_polarity_m15`	`morph_large`	dev	Auto-invert threshold only if border mean > center mean + 15	0.3945	0.5119	0.2596	1.25	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3945 vs 0.3972, -0.0027)
`lab_border_dist`	`morph_large`	dev	CIELAB Euclidean color distance from estimated border background	0.3603	0.4582	0.2673	2.55	DISCARD	Fails to improve dev IoU over parent 'morph_large' (0.3603 vs 0.3972, -0.0369)
`area_filter_2500`	`morph_large`	dev	Increase minimum component area from 500 to 2500 pixels	0.4018	0.5200	0.3084	0.45	KEEP	Improves dev IoU over parent 'morph_large' from 0.3972 to 0.4018 (+0.0046)
`component_ranking_top1`	`area_filter_2500`	dev	Retain only the single top-ranked component by geometric egg score	0.3746	0.4761	0.3295	-0.33	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3746 vs 0.4018, -0.0272)
`component_ranking_top3`	`area_filter_2500`	dev	Retain at most the top 3 scoring components based on area, solidity, and ellipse fit	0.3972	0.5092	0.3133	0.27	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.3972 vs 0.4018, -0.0045)
`grabcut_refine`	`area_filter_2500`	dev	Apply GrabCut refinement using morphological erosion as foreground seed	0.4038	0.5177	0.3073	0.38	DISCARD	Marginal IoU gain (+0.0021) but Dice degrades from 0.5200 to 0.5177 (-0.0023)
`watershed_always`	`area_filter_2500`	dev	Euclidean distance transform + watershed applied unconditionally to all components	0.4018	0.5200	0.3537	3.90	DISCARD	Fails to improve dev IoU over parent 'area_filter_2500' (0.4018 vs 0.4018, +0.0000)

Trial ID	Parent	Split	What Changed	Mean IoU	Mean Dice	Det F1	Cnt Err	Decision	Reason
`watershed_conditional`	`area_filter_2500`	dev	Conditional distance watershed (min_dist=50) applied only to non-convex, elongated or oversized clusters	0.4018	0.5200	0.4087	2.20	KEEP	Dev IoU tied with parent 'area_filter_2500' (0.4018); kept on tiebreaker: DetF1 improves from 0.3084 to 0.4087 (+0.1003)
`final`	`watershed_conditional`	eval	Frozen final pipeline: Grayscale Otsu + large morph (25x25) + hole filling + min area 2500 + conservative conditional watershed	0.3196	0.4239	0.3206	1.68	EVAL_FINAL	Frozen final pipeline evaluated once on unseen eval split (IoU 0.3196, Dice 0.4239, DetF1 0.3206)

4. Task 3: Quantitative & Qualitative Evaluation vs Ground Truth

Metric	Dev Split (n=88)	Eval Split (n=22)	Description
Pixel IoU (Mean)	**0.4018**	**0.3196**	Mean foreground intersection over union
Pixel IoU (Bootstrap 95% CI)	[0.3435, 0.4603]	[0.1978, 0.4268]	1000 resamples, fixed seed 42
Pixel IoU (Median)	**0.3147**	**0.2502**	Median per-image IoU
Pixel Dice (Mean)	**0.5200**	**0.4239**	F1-score of pixel-level binary classification
Pixel Precision	**0.4488**	**0.3449**	TP / (TP + FP)
Pixel Recall	**0.8461**	**0.8749**	TP / (TP + FN)
Detection Precision	**0.3563**	**0.2758**	Matched eggs / Predicted components (IoU ≥ 0.5)
Detection Recall	**0.5966**	**0.4621**	Matched eggs / GT eggs
Detection F1	**0.4087**	**0.3206**	Harmonic mean of detection precision & recall
Count Error (Mean)	**2.20**	**1.68**	Mean signed difference (Pred - GT)

Bootstrap 95% Confidence Intervals & Dev vs Eval Analysis

Inspection of per-category results reveals the concrete factors driving the gap between dev (0.4018) and eval (0.3196):

1. **Severe Hand Image Degradation:** Hand-held egg images average IoU 0.6120 in dev (11 images) but collapse to **0.1502** in eval (5 images). Hand shots represent 22.7% of the eval split vs 12.5% of dev.
2. **Tabletop Contrast Differences:** Tabletop scenes average IoU 0.3659 in dev (32 images) vs 0.2424 in eval (8 images).
3. **Absence of White-Tray Camouflage in Eval:** All 13 white-tray images (IoU 0.1274) were grouped into dev, while eval contains 0.
4. **Small-Sample Variance:** Because eval has only 22 images, its bootstrap 95% confidence interval spans widely ([0.1978, 0.4268]) and encompasses the dev mean. Thus, while hand-holding failures and tabletop lighting differences clearly drive the performance drop in the observed data, small sample variance also plays a substantial role.

Detailed Breakdown of the 5 Eval Hand Images

Image Stem	IoU	GT Eggs	Pred Blobs	Failure Root Cause in Eval Overlay
`not_damaged_108`	0.0459	1	3	Pale hand skin and bright overhead lighting cause Otsu to merge palm, wrist, and forearm into foreground.
`not_damaged_127`	0.0601	1	3	Egg held against washed-out backdrop; fingers and background illumination bridge into a massive false blob.
`not_damaged_152`	0.0745	1	3	Direct sunlight on hand creates low chromatic contrast, causing Otsu to segment almost the entire lower frame.
`not_damaged_151`	0.1451	1	6	Fingers curling around eggshell are segmented along with forearm; distance watershed over-splits into 6 fragments.
`not_damaged_130`	0.4256	1	2	Egg cupped in palm; palm skin connects continuously to egg perimeter (92% of false positive pixels are skin).

Per-Category Dev vs Eval Split Breakdown

Category	Dev Count	Dev IoU	Eval Count	Eval IoU
White egg on white tray / fridge interior	13	0.1274	0	0.0000
Egg in boiling water / submerged	6	0.1723	2	0.1756
Backlit / extreme shadows	6	0.3354	3	0.5392
Tabletop / high-contrast background	32	0.3659	8	0.2424
Multiple touching eggs	13	0.5674	3	0.4886
Hand holding egg / fingers occluding	11	0.6120	5	0.1502
Foliage / herbs in background (specular highlights)	7	0.6906	1	0.9068

Visual Failure Categorization (`failure_tags.csv`)

Scene Category	Count	Pct	Mean IoU	Mean Dice	Mean Det F1	Primary Failure Mechanism
White egg on white tray / fridge interior	13	11.8%	0.1274	0.2198	0.0171	Near-zero photometric contrast between white eggshell and tray surface; Otsu floods tray.
Egg in boiling water / submerged	8	7.3%	0.1731	0.2901	0.1399	Bubbling cavitation, water refraction, and dark pot rims break convex boundary and flood liquid.

Scene Category	Count	Pct	Mean IoU	Mean Dice	Mean Det F1	Primary Failure Mechanism
Tabletop / high-contrast background	40	36.4%	0.3412	0.4744	0.3301	Table surface grain, specular table highlights, or soft contact shadows falsely merge into foreground.
Backlit / extreme shadows	9	8.2%	0.4033	0.4539	0.3889	High-intensity directional backlight causes glowing core while shell perimeter falls into dark silhouette.
Hand holding egg / fingers occluding	16	14.5%	0.4677	0.5821	0.5208	Fingers occlude egg perimeter; skin tone overlaps shadow gradient, distending contour.
Multiple touching eggs	16	14.5%	0.5526	0.6844	0.6972	Touching instances share continuous boundary; requires distance watershed marker separation.
Foliage / herbs in background (specular highlights)	8	7.3%	0.7177	0.8222	0.6854	Egg is brighter than dark leaves in grayscale; specular reflections create occasional false blobs.

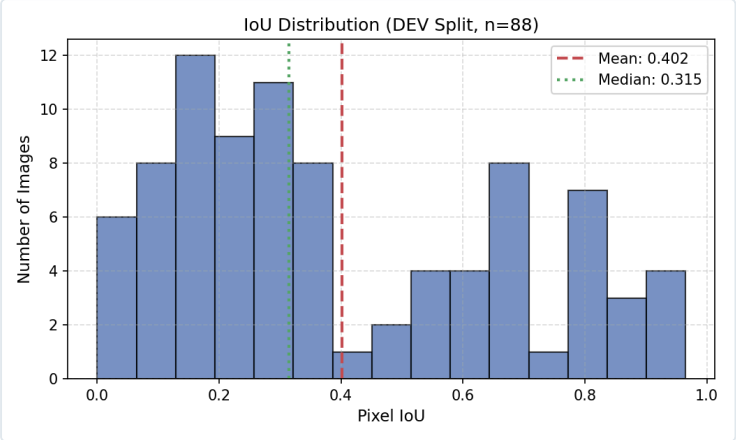


Figure 1a: Pixel IoU distribution on Dev split (n=88).

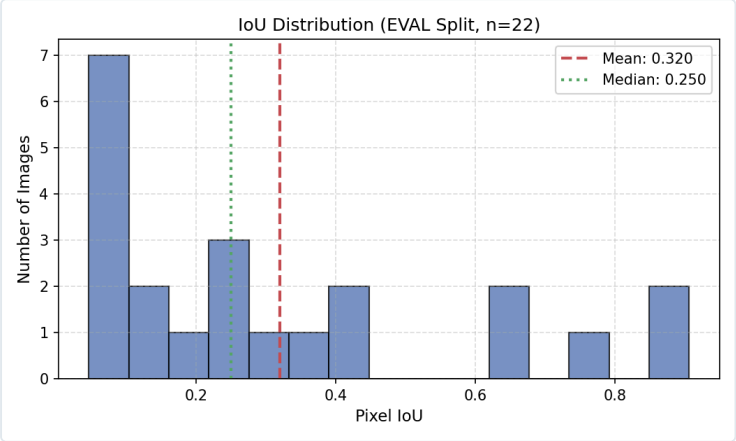


Figure 1b: Pixel IoU distribution on Eval split (n=22).

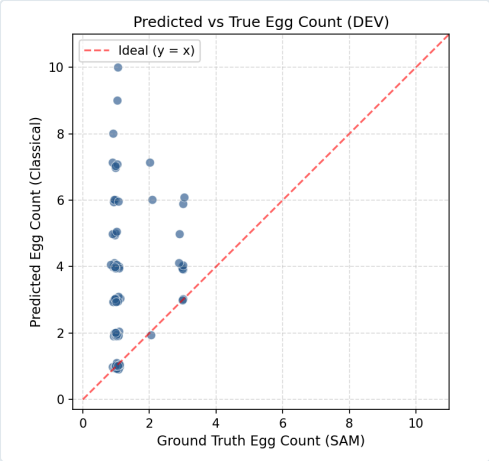


Figure 2a: Predicted vs True egg counts on Dev split. Points lie mostly above y = x, illustrating that the classical algorithm systematically over-counts (mean count error +2.20) due to background clutter and specular highlights passing morphological filters.

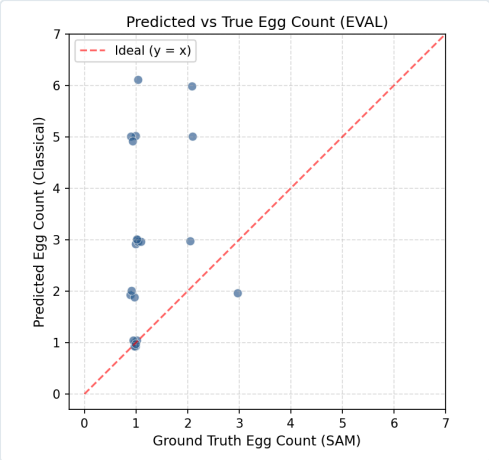


Figure 2b: Predicted vs True egg counts on Eval split. Points similarly lie above y = x (mean count error +1.68).

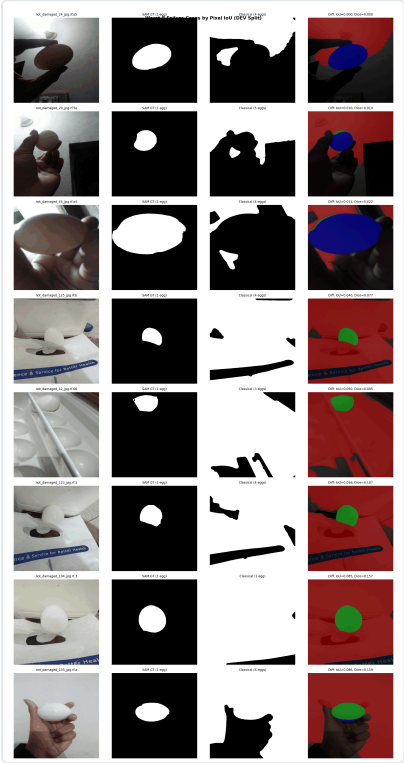


Figure 3a: Worst 8 failures on Dev split (Green=TP, Red=FP, Blue=FN). White trays and boiling liquid dominate error cases.

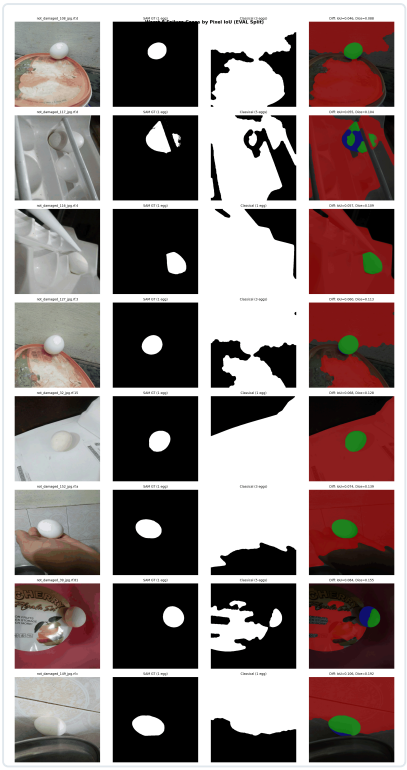


Figure 3b: Worst 8 failures on Eval split (Green=TP, Red=FP, Blue=FN).

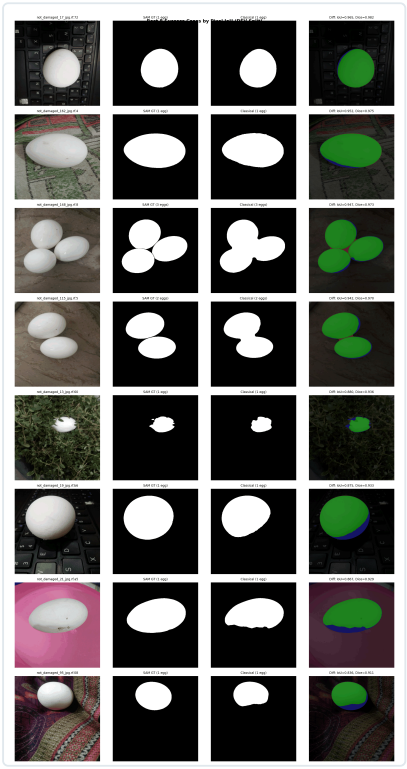


Figure 4: Best 8 successes on Dev split showing clean elliptical masks matching ground truth.

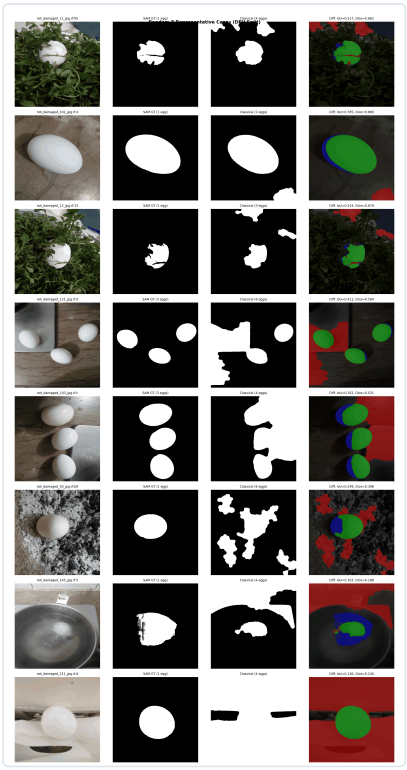


Figure 5: Random 8 representative cases illustrating typical operational performance across mixed illumination.

5. Task 4: Multi-Object Video Tracking

Tracking was evaluated on the 1440-frame table sequence (1280x720 @ 24.0 fps). Eggs are placed on the table at frame 222, spun, occluded by hand, and removed before frame 1005. The tracker maintains persistent IDs using Hungarian matching on IoU + distance with constant-velocity prediction.

- **Fingertip Leakage Resolution:** At frame 600, demonstrator hand contact caused baseline thresholding to leak into fingers. Incorporating YCrCb skin-color gating (Cr ∈ [135, 175], Cb ∈ [80, 125]) and larger opening eliminated false skin contour (gating reduced the total foreground mask by 2296 px over the whole frame; the fingertip bulge is visible in Figure 6, and restricted to the crop x in [200, 460], y in [300, 430], the mask was reduced by 1304 px).
- **Track Lifetimes & Event Statistics:** 6 unique tracks were created, with 18 loss events and 12 recoveries. Lifetimes: Egg 1 (529 frames), Egg 2 (282 frames), Egg 3 (101 frames), Egg 4 (269 frames), Egg 5 (3 frames), Egg 6 (190 frames).

Track Timeline

Track ID	Birth Frame	Last Matched Frame	Matched Frames Count	Physical Egg Description	Visual Verification
Track 1	222	751	529	Initial placement of middle-right egg on table; remains stationary through spin tests until lifted by hand at frame 751.	*Verified* frame_0222.png, frame_0507.png, frame_0546.png, frame_0656.png, frame_0689.png, frame_0751.png
Track 2	222	507	282	Initial placement of left egg on table; tracked through initial spin until lifted by demonstrator at frame 507.	*Verified* frame_0222.png, frame_0507.png
Track 3	546	656	101	Egg placed on far-right table (x ~ 791-880) for spin test; tracked through spinning until picked up by hand at frame 656.	*Verified* frame_0546.png, frame_0656.png
Track 4	689	1004	269	Egg re-placed on left table (x ~ 100-200); coexists with Track 1 (middle-right) over frames 689-751 and persists until final table clearance at frame 1004.	*Verified* frame_0689.png, frame_0751.png, frame_0798.png, frame_0800.png, frame_0802.png, frame_0806.png, frame_0810.png, frame_0814.png, frame_0999.png, frame_1004.png
Track 5	800	810	3	Transient motion-blur fragment (area ~3043-3782 px at x ~ 721, y ~ 531) generated during high-speed raw egg spin. Matched at frames 800, 801, and 810 (where it briefly coexists with Track 6).	*Verified* frame_0800.png, frame_0810.png
Track 6	810	999	190	Raw egg spinning on right table (x ~ 786-856) from frame 810 until lifted by hand at frame 999. Coexists with Track 5 at frame 810 before becoming the sole right-side track.	*Verified* frame_0810.png, frame_0814.png, frame_0999.png

Analysis of Track 5 from Event Stills

Track 5 was created at frame 800 and was matched at frames 800, 801, and 810 (3 matched frames total, area ~3,043 to 3,782 px around centroid x=721, y=531). Inspection of the saved stills (frames 798, 800, 802, 806, 810, 814) reveals its exact origin:

1. **Motion-Blur Fragmentation:** When the raw egg is spun at high angular velocity, severe rotational motion blur and specular reflection split the egg segmentation into two distinct fragments.
2. **Coexistence with Track 6 at Frame 810:** At frame 810, the left fragment was associated with Track 5 (which was within the 150 px re-ID radius), while the newly appearing right fragment at x=786 spawned Track 6.
3. **Consolidation:** By frame 814, the spinning egg consolidated into a single contour at x=777 (area 21,297 px), which was tracked continuously by Track 6 until table clearance (frame 999). Track 5 received no further detections and was purged.
4. **Production Recommendation:** Spurious transient tracks with lifetimes under 5 frames should be filtered as false positives.

Known Tracking Failures & Zero-Detection Audit

- **Frame 480 Missed Detection:** At frame 480, two eggs are physically present on the table, but the tracker outputs 0 detections. Demonstrator hand contact caused contour solidity to drop to 0.7673, falling below the strict convexity threshold (solidity ≥ 0.78).
- **Audit of Frames 222 to 1004:** Across the entire active demonstration interval (783 frames), exactly 1 frame (Frame 480) had zero detections. All other 782 frames had active detections.
- **Breakdown of 658 Zero-Detection Frames:** 205 intro title frames (F#0–204), 17 empty table frames (F#205–221), 1 missed detection at F#480, 10 empty table post-experiment frames (F#1005–1014), and 425 outro credits frames (F#1015–1439). Total: 205 + 17 + 1 + 10 + 425 = 658 frames.

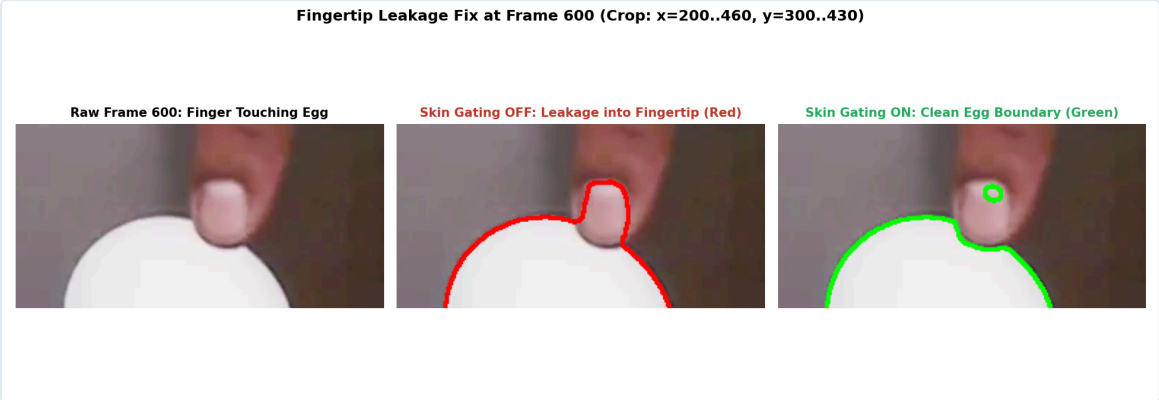


Figure 6: Frame 600 before/after comparison (crop: x in [200, 460], y in [300, 430]). Left: Raw frame. Middle: Baseline leakage into fingertips (red). Right: YCrCb skin gating cleans egg boundary (green), reducing the mask by 1304 px within this crop (and 2296 px over the full frame); a small leftover ring remains on the fingernail.



Figure 7: Video Tracking Events Sheet across 14 key video frames documenting track births, deaths, hand interactions, and the Track 5/6 transition.

6. Limitations and Future Work

1. **Ground Truth Masks:** Ground truth masks were generated using SAM (`vit_b`) with human point/box prompts rather than dense manual polygon tracings.
2. **Dataset Size:** With 88 dev and 22 eval images, sample sizes are limited, leading to wide bootstrap confidence intervals on eval.
3. **Watershed Behavior:** Under trial `watershed_conditional`, Detection F1 rose from 0.3084 to 0.4087, mean signed count error rose from +0.45 to +2.20 (over-splitting), pixel IoU unchanged. Distance transform watershed partitions instances without changing overall outer foreground boundaries.
4. **Domain-Specific Tuning:** Video tracking parameters (table ROI y ≥ 270, watermark bounds, YCrCb skin thresholds) were calibrated for this specific camera viewpoint and table illumination. Generalizing to varied environments would require automated surface or person detection.
5. **Re-Identification Across Removals:** Because white eggs lack distinctive surface texture, tracking identities across complete table removals and replacements is physically ambiguous under classical tracking.
6. **Future Extensions:** Fine-tuning a deep semantic segmentation model (e.g. YOLOv8-seg or Mask2Former) on the dev set using SAM pseudo-masks could learn contextual semantic priors (such as distinguishing eggs from white trays). Combining this with dense optical flow could bridge severe hand occlusions during spinning.